



Advanced Cisco Networking Academy: Designing a Multiprotocol Network with BGP

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Purpose

This lab helps to build a greater understanding of enterprise networking through the configuration of exterior routing protocol BGP connecting different networks using different routing protocols. This lab also helps to build a better understanding of how enterprise networks use exterior routing protocols like BGP to provide redundancy and more reliable and secure connections. Additionally, this lab builds an understanding of how Virtual routers interact with each other to create a complete routing table that goes beyond the specific routing protocol of an area to provide connectivity over a greater area.

Background

BGP or Border Gateway Protocol is the most widely used gateway protocol in the world with virtually the entire world relying on BGP to provide reliable connections between large networks. BGP has 2 iterations, Internal BGP and External BGP, iBGP is used to route within the same AS (Autonomous System: Large Network) while in this lab eBGP is used to route between separate Autonomous Systems allowing connectivity between different networks using unique routing protocols. This reflects how BGP provides connectivity not just to large corporations who need to use different routing protocols to fit the network's needs but also reflects how the BGP provides connection to the whole internet. Because BGP is the routing protocol that serves as the glue that holds together countless networks letting the internet work, it has several built-in redundancy protocols that are configurable within the same router without the need of backup routers to provide that redundancy. This redundancy is extremely important when regards to ensuring connection between BGP, there have been several cases where BGP was slightly misconfigured without a backup configuration to take place leading to major security vulnerabilities and wide area network outages. A notable incident of a BGP failure is the Rogers Communication Outage of July 2022, a failure in the network caused wide-spread and long-lasting service outages through Canada preventing essential services like bank transfers, home internet, and electronic payment systems. This outage affected an estimated amount of 12 million people in Canada, which is approximately 30% of Canada's population, the halt of all electronic payments and fiscal transfers significantly slowed the Canadian economy causing an estimated damage of \$142 million. There are many more incidents where BGP failure has caused significant damage to Companies, People, and Governments. Incidents like when a route leak caused US domestic traffic to be routed to China, Facebook Outage causing all Facebook services to go down, and many more. Incidents like these significantly emphasize the importance of not just configuring BGP to provide connection to the internet, but also the importance of

configuring redundancy, reliability, and security all within BGP to ensure essential services and be reached and provided.

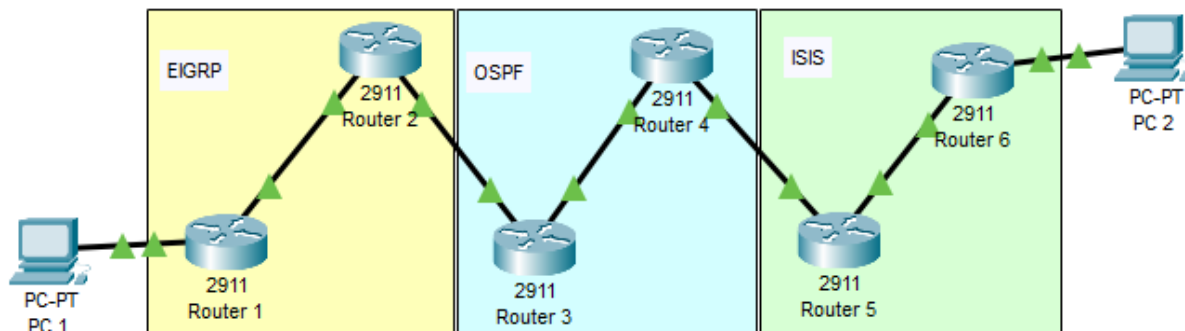
BGP operates on the application level with protocols like OSPF, HTTPS, and DNS. It operates in port 179 using TCP and sends keep-alive messages every 30 seconds to maintain connection between its neighbors. BGP is a very flexible protocol in that it can run multiple topologies at the same time, meaning that it can accommodate the needs of two networks simultaneously through BGP Multi-Instance creating multiple separate processes of BGP with separate configurations for each network, this is configured through having multiple different autonomous systems on the same router. For a BGP-enabled router to establish a peer connection, it needs to go through six states: Idle, Connect, Active, OpenSent, OpenConfirm, Established. The Idle State is the first state the BGP router starts in; it rejects all inbound BGP connection attempts for security and attempts to establish a connection with the configured peer. This peer does not reject the connection because they are both manually configured as peers to each other. After a TCP message is sent, the BGP state transitions to the connect state. If the TCP connection fails the BGP state changes to active, a retry timer starts and attempts to establish a connection again. Once the router is in the Connect state, it waits until the TCP connection is fully established then transitions to the OpenSent state. Once the router is in the OpenSent state, it sends an Open message which contains information about the BGP router such as the Version, AS number, and a 32-bit Identifier and then waits to receive one in return. Once both routers receive an OpenSent message, they transition to the OpenConfirm State, and after both sides send keepalive messages, they will both transition to the established state where routing information will be exchanged. Understanding how each state works and the requirements to achieve and transition to the next state is extremely useful information when trying to debug configurations in BGP, knowing that the routers are stuck in one state will directly point out and narrow the number of things to fix or check to establish vital internetwork connections.

Lab Summary

In this lab, I first configured 3 separate autonomous systems on IPv4 EIGRP, OSPF, IS-IS, and IPv6 EIGRP and OSPF. Each of these areas had 2 routers sharing the same protocol and ASN making for a total of 3 routers. Once the Internal Routing Protocols were properly configured and established, I configured BGP between the routing domains to establish connectivity between the 2 border routers in each domain. Then, to establish connectivity between all routers, configure each routing protocol to redistribute other routing protocols in its routing table so that other routers can get those needed routes. After establishing connectivity between all routers, I configured

Graceful Restart and Stateful Switch Over Route Refreshers which allows peers to readvertise configurations without a hard reset.

Network Diagram



Router 1 (EIGRP)	G/0/0 (PC1)	G0/0/1 (R2 G0/0/1)
	192.168.1.1	192.168.2.1
Router 2	G/0/0 (R3 0/0/1)	G0/0/1 (R1 G0/0/1)
	192.168.3.1	192.168.2.2
Router 3 (OSPF)	G/0/0 (R4 0/0/1)	G0/0/1 (R2 0/0/1)
	192.168.4.1	192.168.3.2
Router 4	G/0/0 (R5 0/0/1)	G0/0/1 (R3 0/0/0)
	192.168.5.1	192.168. 4.2
Router 5 (ISIS)	G/0/0 (R6 0/0/1)	G0/0/1 (R4 0/0/0)
	192.168.6.1	192.168.5.2
Router 6	G/0/0 (PC 2)	G0/0/1 (R5 0/0/0)
	192.168. 7.1	192.168. 6.2

Lab Commands

router bgp (ASN): This creates a BGP process with the AS number inputted in the command.

Neighbor (ipv4/ipv6 address) remote-as (neighbor ASN): This configures the bgp router to establish neighbors

Address-family ipv4: This command allows for configuration of ipv4 neighbor activation and route redistribution

Neighbor (ipv4/ipv6) activate: This initiates a BGP session between established neighbors and is done in the address-family sublevel

Redistribute (Protocol/Connected) (ASN): By using the redistribution command different processes on each router can distribute their routes to other routers allowing for all routes to converge into the routing table so that they establish connectivity throughout the whole network. This command sometimes needs to include metrics by adding “metric” to the end of the command and then adding the bandwidth of the link between the routers. This metric is necessary to establish a connection as other routing protocols need the metric information to avoid routing loops and properly calculate the cost of each link between the routers.

bgp graceful-restart: This command allows for BGP connections to stay active and established while a router is being restarted, this is especially helpful when a configuration update/change is needed but the router still need to be active providing connections to its users.

bgp sso route-refresh-enable: Similar to graceful-restart this command lets a BGP router to refresh new routes in the event of a Stateful Switch Over without needing to tear down the BGP session which allows for the BGP router to provide the connection in the event of a failure without needing to compromise the networks' reliability.

Configurations

Router 1:

R1#show run

Building configuration...

Current configuration : 2114 bytes

!

! Last configuration change at 21:37:13 UTC Mon Nov 17 2025

!

version 15.5

service timestamps debug datetime msec

service timestamps log datetime msec

no platform punt-keepalive disable-kernel-core

!

hostname R1

!

boot-start-marker

```
boot-end-marker
!
!
vrf definition Mgmt-intf
!
address-family ipv4
exit-address-family
!
address-family ipv6
exit-address-family
!
!
no aaa new-model
!
!
!
!
!
!
!
!
!
!
!

!
!
!
ipv6 unicast-routing
!
!
!
!
!
!
!
```

```
subscriber templating
vtp domain cisco
vtp mode transparent
multilink bundle-name authenticated
!
!
!
!
license udi pid ISR4321/K9 sn FDO214421BY
!
spanning-tree extend system-id
!
!
redundancy
mode none
!
!
vlan internal allocation policy ascending
!
vlan 10
name Guest
!
vlan 20
name Home
!
vlan 30
name Enterprise
!
!
!
!
!
!
interface GigabitEthernet0/0/0
ip address 192.168.1.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:1::1/64
ipv6 enable
```

```
ipv6 eigrp 30
!
interface GigabitEthernet0/0/1
ip address 192.168.2.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:2::1/64
ipv6 enable
ipv6 eigrp 30
!
interface Serial0/1/0
no ip address
shutdown
!
interface Serial0/1/1
no ip address
shutdown
!
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0/2/1
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
interface Vlan1
no ip address
shutdown
!
!
```



```
router eigrp LAB
!
address-family ipv4 unicast autonomous-system 10
!
  topology base
    redistribute connected
  exit-af-topology
  network 0.0.0.0
  eigrp router-id 1.1.1.1
exit-address-family
!
address-family ipv6 unicast autonomous-system 10
!
  topology base
    redistribute connected
  exit-af-topology
  eigrp router-id 1.1.1.1
exit-address-family
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router eigrp 30
  eigrp router-id 1.1.1.1
!
!
!
!
control-plane
!
!
line con 0
  stopbits 1
line aux 0
  stopbits 1
```

```
line vty 0 4
```

```
login
```

```
!
```

```
!
```

```
End
```

```
R1#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.1.0/24 is directly connected, GigabitEthernet0/0/0

L 192.168.1.1/32 is directly connected, GigabitEthernet0/0/0

192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.2.0/24 is directly connected, GigabitEthernet0/0/1

L 192.168.2.1/32 is directly connected, GigabitEthernet0/0/1

D 192.168.3.0/24 [90/15360] via 192.168.2.2, 2d23h, GigabitEthernet0/0/1

D EX 192.168.4.0/24

[170/1029120] via 192.168.2.2, 2d23h, GigabitEthernet0/0/1

D EX 192.168.5.0/24

[170/1029120] via 192.168.2.2, 2d23h, GigabitEthernet0/0/1

D EX 192.168.6.0/24

[170/1029120] via 192.168.2.2, 2d23h, GigabitEthernet0/0/1

D EX 192.168.7.0/24

[170/1029120] via 192.168.2.2, 02:11:13, GigabitEthernet0/0/1

D EX 192.168.10.0/24

[170/1029120] via 192.168.2.2, 2d23h, GigabitEthernet0/0/1

D EX 192.168.11.0/24

[170/1029120] via 192.168.2.2, 2d23h, GigabitEthernet0/0/1

```
R1#show ipv6 route
```

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

C 2001:DB8:ACAD:1::/64 [0/0]

via GigabitEthernet0/0/0, directly connected

L 2001:DB8:ACAD:1::1/128 [0/0]

via GigabitEthernet0/0/0, receive

C 2001:DB8:ACAD:2::/64 [0/0]

via GigabitEthernet0/0/1, directly connected

L 2001:DB8:ACAD:2::1/128 [0/0]

via GigabitEthernet0/0/1, receive

D 2001:DB8:ACAD:3::/64 [90/3072]

via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/1

EX 2001:DB8:ACAD:4::/64 [170/281856]

via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/1

EX 2001:DB8:ACAD:5::/64 [170/281856]

via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/1

EX 2001:DB8:ACAD:6::/64 [170/281856]

via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/1

EX 2001:DB8:ACAD:7::/64 [170/281856]

via FE80::B6A8:B9FF:FE47:92C1, GigabitEthernet0/0/1

L FF00::/8 [0/0]

via Null0, receive

Router 2:

R2#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

D 192.168.1.0/24
[90/15360] via 192.168.2.1, 02:03:59, GigabitEthernet0/0/1
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.2.0/24 is directly connected, GigabitEthernet0/0/1
L 192.168.2.2/32 is directly connected, GigabitEthernet0/0/1
192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.3.0/24 is directly connected, GigabitEthernet0/0/0
L 192.168.3.1/32 is directly connected, GigabitEthernet0/0/0
B 192.168.4.0/24 [20/0] via 192.168.3.2, 2d23h
B 192.168.5.0/24 [20/10000] via 192.168.3.2, 2d23h
B 192.168.6.0/24 [20/10000] via 192.168.3.2, 2d23h
B 192.168.7.0/24 [20/10000] via 192.168.3.2, 02:02:08
B 192.168.10.0/24 [20/10000] via 192.168.3.2, 2d23h
B 192.168.11.0/24 [20/10000] via 192.168.3.2, 2d23h

R2#show ipv6 route

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, H - NHRP, I1 - ISIS L1

I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP

EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination

NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1

OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

a - Application

D 2001:DB8:ACAD:1::/64 [90/3072]
via FE80::B6A8:B9FF:FE01:B511, GigabitEthernet0/0/1
C 2001:DB8:ACAD:2::/64 [0/0]
via GigabitEthernet0/0/1, directly connected
L 2001:DB8:ACAD:2::2/128 [0/0]
via GigabitEthernet0/0/1, receive
C 2001:DB8:ACAD:3::/64 [0/0]
via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:3::1/128 [0/0]
via GigabitEthernet0/0/0, receive
B 2001:DB8:ACAD:4::/64 [20/0]

```

    via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
B 2001:DB8:ACAD:5::/64 [20/10000]
    via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
B 2001:DB8:ACAD:6::/64 [20/10000]
    via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
B 2001:DB8:ACAD:7::/64 [20/10000]
    via FE80::B6A8:B9FF:FE01:AE51, GigabitEthernet0/0/0
L FF00::/8 [0/0]
    via Null0, receive

```

R2#show ip bgp

BGP table version is 40, local router ID is 192.168.3.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 192.168.1.0	192.168.2.1	15360		32768	?
*> 192.168.2.0	0.0.0.0	0		32768	i
* 192.168.3.0	192.168.3.2	0		0 2	i
*> 0.0.0.0	0	32768			i
*> 192.168.4.0	192.168.3.2	0		0 2	?
*> 192.168.5.0	192.168.3.2	10000		0 2	?
*> 192.168.6.0	192.168.3.2	10000		0 2	?
*> 192.168.7.0	192.168.3.2	10000		0 2	?
*> 192.168.10.0	192.168.3.2	10000		0 2	?
*> 192.168.11.0	192.168.3.2	10000		0 2	?

R2#show bgp neighbors

BGP neighbor is 192.168.3.2, remote AS 2, external link

BGP version 4, remote router ID 192.168.4.1

BGP state = Established, up for 3d00h

Last read 00:00:07, last write 00:00:42, hold time is 180, keepalive interval is 60 seconds

Neighbor sessions:

1 active, is not multisession capable (disabled)

Neighbor capabilities:

Route refresh: advertised and received(new)

Four-octets ASN Capability: advertised and received
Address family IPv4 Unicast: advertised and received
Enhanced Refresh Capability: advertised and received
Multisession Capability:

Stateful switchover support enabled: NO for session 1

Message statistics:

InQ depth is 0

OutQ depth is 0

	Sent	Rcvd
Opens:	1	1
Notifications:	0	0
Updates:	15	8
Keepalives:	4760	4764
Route Refresh:	0	0
Total:	4776	4773

Do log neighbor state changes (via global configuration)

Default minimum time between advertisement runs is 30 seconds

For address family: IPv4 Unicast

Session: 192.168.3.2

BGP table version 40, neighbor version 40/0

Output queue size : 0

Index 1, Advertise bit 0

1 update-group member

Slow-peer detection is disabled

Slow-peer split-update-group dynamic is disabled

	Sent	Rcvd
Prefix activity:	----	----
Prefixes Current:	3	7 (Consumes 840 bytes)
Prefixes Total:	11	9
Implicit Withdraw:	5	0
Explicit Withdraw:	3	2
Used as bestpath:	n/a	6
Used as multipath:	n/a	0

	Outbound	Inbound
--	----------	---------

Local Policy Denied Prefixes:	-----	-----
-------------------------------	-------	-------

Bestpath from this peer: 8 n/a

Total: 8 0

Number of NLRIs in the update sent: max 1, min 0

Last detected as dynamic slow peer: never

Dynamic slow peer recovered: never

Refresh Epoch: 1

Last Sent Refresh Start-of-rib: never

Last Sent Refresh End-of-rib: never

Last Received Refresh Start-of-rib: never

Last Received Refresh End-of-rib: never

	Sent	Rcvd
Refresh activity:	----	----
Refresh Start-of-RIB	0	0
Refresh End-of-RIB	0	0

Address tracking is enabled, the RIB does have a route to 192.168.3.2

Connections established 1; dropped 0

Last reset never

External BGP neighbor configured for connected checks (single-hop no-disable-connected-check)

Interface associated: GigabitEthernet0/0/0 (peering address in same link)

Transport(tcp) path-mtu-discovery is enabled

Graceful-Restart is disabled

SSO is disabled

Connection state is ESTAB, I/O status: 1, unread input bytes: 0

Connection is ECN Disabled, Mininum incoming TTL 0, Outgoing TTL 1

Local host: 192.168.3.1, Local port: 55694

Foreign host: 192.168.3.2, Foreign port: 179

Connection tableid (VRF): 0

Maximum output segment queue size: 50

Enqueued packets for retransmit: 0, input: 0 mis-ordered: 0 (0 bytes)

Event Timers (current time is 0x1B64B02A):

Timer	Starts	Wakeups	Next
Retrans	4775	1	0x0
TimeWait	0	0	0x0
AckHold	4771	4704	0x0

SendWnd	0	0	0x0
KeepAlive	0	0	0x0
GiveUp	0	0	0x0
PmtuAger	258455	258454	0x1B64B2AA
DeadWait	0	0	0x0
Linger	0	0	0x0
ProcessQ	0	0	0x0

iss: 880031668 snduna: 880122864 sndnxt: 880122864

irs: 543734261 rcvnxt: 543825198

sndwnd: 15928 scale: 0 maxrcvwnd: 16384

rcvwnd: 16156 scale: 0 delrcvwnd: 228

SRTT: 1000 ms, RTTO: 1003 ms, RTV: 3 ms, KRTT: 0 ms

minRTT: 0 ms, maxRTT: 1000 ms, ACK hold: 200 ms

uptime: 260143518 ms, Sent idletime: 7085 ms, Receive idletime: 7286 ms

Status Flags: active open

Option Flags: nagle, path mtu capable

IP Precedence value : 6

Datagrams (max data segment is 1460 bytes):

Rcvd: 9539 (out of order: 0), with data: 4772, total data bytes: 90936

Sent: 9544 (retransmit: 1, fastretransmit: 0, partialack: 0, Second Congestion: 0), with data: 4774, total data bytes: 91195

Packets received in fast path: 0, fast processed: 0, slow path: 0

fast lock acquisition failures: 0, slow path: 0

TCP Semaphore 0x7FD22285CE00 FREE

Router 3:

R3#show run

Building configuration...

Current configuration : 2670 bytes

!

! Last configuration change at 22:12:23 UTC Mon Nov 17 2025

!

version 15.5

service timestamps debug datetime msec

service timestamps log datetime msec

no platform punt-keepalive disable-kernel-core

!

hostname R3

!

boot-start-marker

boot-end-marker

!

!

vrf definition Mgmt-intf

!

address-family ipv4

exit-address-family

!

address-family ipv6

exit-address-family

!

!

no aaa new-model

!

!

!

!

!

!

!

!

!

!

!

!

!
!
ipv6 unicast-routing
!
!
!
!
!
!
!
subscriber templating
vtp domain cisco
vtp mode transparent
!
multilink bundle-name authenticated
!
!
!
!
license udi pid ISR4321/K9 sn FDO214420HY
license boot level securityk9
!
spanning-tree extend system-id
!
!
redundancy
mode none
!
!
!
!
!
vlan internal allocation policy ascending
!
vlan 10,20
!
!
!

!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!

```
interface GigabitEthernet0/0/0
ip address 192.168.4.1 255.255.255.0
ip ospf 1 area 0
negotiation auto
ipv6 address 2001:DB8:ACAD:4::1/64
ipv6 enable
ipv6 ospf 10 area 0
!
```

```
interface GigabitEthernet0/0/1
ip address 192.168.3.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:3::2/64
ipv6 enable
ipv6 ospf 10 area 0
!
```

```
interface Serial0/1/0
no ip address
!
```

```
interface Serial0/1/1
```

```
no ip address
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
!
interface Vlan1
no ip address
!
router ospfv3 1
!
address-family ipv6 unicast
  redistribute bgp 2 metric 10000
exit-address-family
!
router ospf 1
router-id 3.3.3.3
redistribute connected subnets
redistribute bgp 2 subnets
redistribute isis level-2 subnets
redistribute eigrp 10
!
router isis
!
router bgp 2
bgp log-neighbor-changes
neighbor 2001:DB8:ACAD:3::1 remote-as 1
neighbor 192.168.3.1 remote-as 1
!
address-family ipv4
  network 192.168.3.0
  redistribute connected
  redistribute ospf 1 metric 10000 match internal external 1 external 2
  redistribute isis level-2
  redistribute eigrp 10
  no neighbor 2001:DB8:ACAD:3::1 activate
  neighbor 192.168.3.1 activate
```

```
exit-address-family
!
address-family ipv6
  redistribute connected
  redistribute ospf 1 metric 10000 match internal external 1 external 2 include-
connected
  redistribute ospf 10 metric 10000 match internal external 1 external 2 include-
connected
  redistribute eigrp 20
  redistribute eigrp 10
  neighbor 2001:DB8:ACAD:3::1 activate
exit-address-family
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router ospf 10
router-id 3.3.3.3
redistribute connected
redistribute bgp 2 metric 10000
!
!
!
!
control-plane
!
!
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
!
!
```

end

R3#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

B 192.168.1.0/24 [20/15360] via 192.168.3.1, 02:22:56

B 192.168.2.0/24 [20/0] via 192.168.3.1, 3d00h

192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.3.0/24 is directly connected, GigabitEthernet0/0/1

L 192.168.3.2/32 is directly connected, GigabitEthernet0/0/1

192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.4.0/24 is directly connected, GigabitEthernet0/0/0

L 192.168.4.1/32 is directly connected, GigabitEthernet0/0/0

O E2 192.168.5.0/24 [110/20] via 192.168.4.2, 3d00h, GigabitEthernet0/0/0

O E2 192.168.6.0/24 [110/1] via 192.168.4.2, 3d00h, GigabitEthernet0/0/0

O E2 192.168.7.0/24 [110/1] via 192.168.4.2, 02:21:23, GigabitEthernet0/0/0

O E2 192.168.10.0/24 [110/1] via 192.168.4.2, 3d00h, GigabitEthernet0/0/0

O E2 192.168.11.0/24 [110/1] via 192.168.4.2, 3d00h, GigabitEthernet0/0/0

R3#show ipv6 route

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, H - NHRP, I1 - ISIS L1

I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP

EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination

NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1

OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

a - Application

B 2001:DB8:ACAD:1::/64 [20/10000]

via FE80::B6A8:B9FF:FE47:92C0, GigabitEthernet0/0/1

B 2001:DB8:ACAD:2::/64 [20/0]
 via FE80::B6A8:B9FF:FE47:92C0, GigabitEthernet0/0/1

C 2001:DB8:ACAD:3::/64 [0/0]
 via GigabitEthernet0/0/1, directly connected

L 2001:DB8:ACAD:3::2/128 [0/0]
 via GigabitEthernet0/0/1, receive

C 2001:DB8:ACAD:4::/64 [0/0]
 via GigabitEthernet0/0/0, directly connected

L 2001:DB8:ACAD:4::1/128 [0/0]
 via GigabitEthernet0/0/0, receive

O 2001:DB8:ACAD:5::/64 [110/2]
 via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0

OE2 2001:DB8:ACAD:6::/64 [110/10000]
 via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0

OE2 2001:DB8:ACAD:7::/64 [110/10000]
 via FE80::521C:B0FF:FE2D:7101, GigabitEthernet0/0/0

L FF00::/8 [0/0]
 via Null0, receive

R3#show bgp neighbor

BGP neighbor is 192.168.3.1, remote AS 1, external link

BGP version 4, remote router ID 192.168.3.1

BGP state = Established, up for 3d00h

Last read 00:00:31, last write 00:00:45, hold time is 180, keepalive interval is 60 seconds

Neighbor sessions:

1 active, is not multisession capable (disabled)

Neighbor capabilities:

Route refresh: advertised and received(new)

Four-octets ASN Capability: advertised and received

Address family IPv4 Unicast: advertised and received

Enhanced Refresh Capability: advertised and received

Multisession Capability:

Stateful switchover support enabled: NO for session 1

Message statistics:

InQ depth is 0

OutQ depth is 0

Sent Rcvd

Opens:	1	1
Notifications:	0	0
Updates:	8	15
Keepalives:	4751	4748
Route Refresh:	0	0
Total:	4760	4764

Do log neighbor state changes (via global configuration)

Default minimum time between advertisement runs is 30 seconds

For address family: IPv4 Unicast

Session: 192.168.3.1

BGP table version 22, neighbor version 22/0

Output queue size : 0

Index 1, Advertise bit 0

1 update-group member

Slow-peer detection is disabled

Slow-peer split-update-group dynamic is disabled

	Sent	Rcvd
Prefix activity:	----	----
Prefixes Current:	7	3 (Consumes 360 bytes)
Prefixes Total:	9	11
Implicit Withdraw:	0	5
Explicit Withdraw:	2	3
Used as bestpath:	n/a	2
Used as multipath:	n/a	0

	Outbound	Inbound
Local Policy Denied Prefixes:	-----	-----
Bestpath from this peer:	7	n/a
Total:	7	0

Number of NLRIs in the update sent: max 5, min 0

Last detected as dynamic slow peer: never

Dynamic slow peer recovered: never

Refresh Epoch: 1

Last Sent Refresh Start-of-rib: never

Last Sent Refresh End-of-rib: never

Last Received Refresh Start-of-rib: never

Last Received Refresh End-of-rib: never

	Sent	Rcvd
Refresh activity:	----	----
Refresh Start-of-RIB	0	0
Refresh End-of-RIB	0	0

Address tracking is enabled, the RIB does have a route to 192.168.3.1

Connections established 1; dropped 0

Last reset never

External BGP neighbor configured for connected checks (single-hop no-disable-connected-check)

Interface associated: GigabitEthernet0/0/1 (peering address in same link)

Transport(tcp) path-mtu-discovery is enabled

Graceful-Restart is disabled

SSO is disabled

Connection state is ESTAB, I/O status: 1, unread input bytes: 0

Connection is ECN Disabled, Mininum incoming TTL 0, Outgoing TTL 1

Local host: 192.168.3.2, Local port: 179

Foreign host: 192.168.3.1, Foreign port: 55694

Connection tableid (VRF): 0

Maximum output segment queue size: 50

Enqueued packets for retransmit: 0, input: 0 mis-ordered: 0 (0 bytes)

Event Timers (current time is 0x1B5AAF8F):

Timer	Starts	Wakeups	Next
Retrans	4758	0	0x0
TimeWait	0	0	0x0
AckHold	4761	4690	0x0
SendWnd	0	0	0x0
KeepAlive	0	0	0x0
GiveUp	0	0	0x0
PmtuAger	0	0	0x0
DeadWait	0	0	0x0
Linger	0	0	0x0
ProcessQ	0	0	0x0

iss: 543734261 snduna: 543824951 sndnxt: 543824951

irs: 880031668 rcvnxt: 880122636

sndwnd: 14940 scale: 0 maxrcvwnd: 16384
rcvwnd: 16156 scale: 0 delrcvwnd: 228

SRTT: 1000 ms, RTTO: 1003 ms, RTV: 3 ms, KRTT: 0 ms
minRTT: 0 ms, maxRTT: 1000 ms, ACK hold: 200 ms
uptime: 259484600 ms, Sent idletime: 31000 ms, Receive idletime: 31200 ms
Status Flags: passive open, gen tcbs
Option Flags: nagle, path mtu capable
IP Precedence value : 6

Datagrams (max data segment is 1460 bytes):

Rcvd: 9519 (out of order: 0), with data: 4762, total data bytes: 90967

Sent: 9514 (retransmit: 0, fastretransmit: 0, partialack: 0, Second Congestion: 0), with data: 4759, total data bytes: 90689

Packets received in fast path: 0, fast processed: 0, slow path: 0

fast lock acquisition failures: 0, slow path: 0

TCP Semaphore 0x7F1567830630 FREE

R3#show ip bgp

BGP table version is 22, local router ID is 192.168.4.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 192.168.1.0	192.168.3.1	15360		0 1 ?	
*> 192.168.2.0	192.168.3.1	0		0 1 i	
* 192.168.3.0	192.168.3.1	0		0 1 i	
*> 0.0.0.0	0	32768		i	
*> 192.168.4.0	0.0.0.0	0		32768 ?	
*> 192.168.5.0	192.168.4.2	10000		32768 ?	
*> 192.168.6.0	192.168.4.2	10000		32768 ?	
*> 192.168.7.0	192.168.4.2	10000		32768 ?	
*> 192.168.10.0	192.168.4.2	10000		32768 ?	
*> 192.168.11.0	192.168.4.2	10000		32768 ?	

Router 4:

R4#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O E2 192.168.1.0/24 [110/1] via 192.168.4.1, 02:37:20, GigabitEthernet0/0/1

O E2 192.168.2.0/24 [110/1] via 192.168.4.1, 3d00h, GigabitEthernet0/0/1

O E2 192.168.3.0/24 [110/20] via 192.168.4.1, 3d00h, GigabitEthernet0/0/1

192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.4.0/24 is directly connected, GigabitEthernet0/0/1

L 192.168.4.2/32 is directly connected, GigabitEthernet0/0/1

192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.5.0/24 is directly connected, GigabitEthernet0/0/0

L 192.168.5.1/32 is directly connected, GigabitEthernet0/0/0

B 192.168.6.0/24 [20/0] via 192.168.5.2, 3d00h

B 192.168.7.0/24 [20/10000] via 192.168.5.2, 02:35:47

B 192.168.10.0/24 [20/0] via 192.168.5.2, 3d00h

B 192.168.11.0/24 [20/10000] via 192.168.5.2, 3d00h

R4#show ipv6 route

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, H - NHRP, I1 - ISIS L1

I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP

EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination

NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1

OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

a - Application

OE2 2001:DB8:ACAD:1::/64 [110/10000]

via FE80::B6A8:B9FF:FE01:AE50, GigabitEthernet0/0/1
 OE2 2001:DB8:ACAD:2::/64 [110/10000]
 via FE80::B6A8:B9FF:FE01:AE50, GigabitEthernet0/0/1
 O 2001:DB8:ACAD:3::/64 [110/2]
 via FE80::B6A8:B9FF:FE01:AE50, GigabitEthernet0/0/1
 C 2001:DB8:ACAD:4::/64 [0/0]
 via GigabitEthernet0/0/1, directly connected
 L 2001:DB8:ACAD:4::2/128 [0/0]
 via GigabitEthernet0/0/1, receive
 C 2001:DB8:ACAD:5::/64 [0/0]
 via GigabitEthernet0/0/0, directly connected
 L 2001:DB8:ACAD:5::1/128 [0/0]
 via GigabitEthernet0/0/0, receive
 B 2001:DB8:ACAD:6::/64 [20/0]
 via FE80::6EDD:30FF:FEB6:6031, GigabitEthernet0/0/0
 B 2001:DB8:ACAD:7::/64 [20/3072]
 via FE80::6EDD:30FF:FEB6:6031, GigabitEthernet0/0/0
 L FF00::/8 [0/0]
 via Null0, receive

R4#show ip bgp

BGP table version is 20, local router ID is 192.168.5.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 192.168.1.0	192.168.4.1	10000		32768	?
*> 192.168.2.0	192.168.4.1	10000		32768	?
*> 192.168.3.0	192.168.4.1	10000		32768	?
*> 192.168.4.0	0.0.0.0	0		32768	?
* 192.168.5.0	192.168.5.2	0		0 4	?
*> 0.0.0.0	0	32768			?
*> 192.168.6.0	192.168.5.2	0		0 4	?
*> 192.168.7.0	192.168.5.2	10000		0 4	?
*> 192.168.10.0	192.168.5.2	0		0 4	?
*> 192.168.11.0	192.168.5.2	10000		0 4	?

R4#show bgp neighbors

BGP neighbor is 192.168.5.2, remote AS 4, external link

BGP version 4, remote router ID 192.168.10.1

BGP state = Established, up for 3d00h

Last read 00:00:02, last write 00:00:09, hold time is 180, keepalive interval is 60 seconds

Neighbor sessions:

1 active, is not multisession capable (disabled)

Neighbor capabilities:

Route refresh: advertised and received(new)

Four-octets ASN Capability: advertised and received

Address family IPv4 Unicast: advertised and received

Graceful Restart Capability: advertised

Enhanced Refresh Capability: advertised and received

Multisession Capability:

Stateful switchover support enabled: NO for session 1

Message statistics:

InQ depth is 0

OutQ depth is 0

	Sent	Rcvd
Opens:	1	1
Notifications:	0	0
Updates:	12	13
Keepalives:	4770	4773
Route Refresh:	1	0
Total:	4786	4789

Do log neighbor state changes (via global configuration)

Default minimum time between advertisement runs is 30 seconds

For address family: IPv4 Unicast

Session: 192.168.5.2

BGP table version 20, neighbor version 20/0

Output queue size : 0

Index 1, Advertise bit 0

1 update-group member

Slow-peer detection is disabled

Slow-peer split-update-group dynamic is disabled

	Sent	Rcvd
Prefix activity:	----	----
Prefixes Current:	5	5 (Consumes 600 bytes)
Prefixes Total:	13	16
Implicit Withdraw:	5	9
Explicit Withdraw:	3	2
Used as bestpath:	n/a	4
Used as multipath:	n/a	0

	Outbound	Inbound
Local Policy Denied Prefixes:	-----	-----
Bestpath from this peer:	10	n/a
Total:	10	0

Number of NLRIs in the update sent: max 3, min 0
 Last detected as dynamic slow peer: never
 Dynamic slow peer recovered: never
 Refresh Epoch: 2
 Last Sent Refresh Start-of-rib: 00:53:33
 Last Sent Refresh End-of-rib: 00:53:33
 Refresh-Out took 0 seconds
 Last Received Refresh Start-of-rib: 00:53:32
 Last Received Refresh End-of-rib: 00:53:32
 Refresh-In took 0 seconds

	Sent	Rcvd
Refresh activity:	----	----
Refresh Start-of-RIB	1	1
Refresh End-of-RIB	1	1

Address tracking is enabled, the RIB does have a route to 192.168.5.2
 Connections established 1; dropped 0
 Last reset never
 External BGP neighbor configured for connected checks (single-hop no-disable-connected-check)
 Interface associated: GigabitEthernet0/0/0 (peering address in same link)
 Transport(tcp) path-mtu-discovery is enabled
 Graceful-Restart is enabled, restart-time 120 seconds, stalepath-time 360 seconds
 SSO is disabled
 Connection state is ESTAB, I/O status: 1, unread input bytes: 0

Connection is ECN Disabled, Minimum incoming TTL 0, Outgoing TTL 1
Local host: 192.168.5.1, Local port: 45641
Foreign host: 192.168.5.2, Foreign port: 179
Connection tableid (VRF): 0
Maximum output segment queue size: 50

Enqueued packets for retransmit: 0, input: 0 mis-ordered: 0 (0 bytes)

Event Timers (current time is 0x1B675FA1):

Timer	Starts	Wakeup	Next
Retrans	4783	1	0x0
TimeWait	0	0	0x0
AckHold	4783	4700	0x0
SendWnd	0	0	0x0
KeepAlive	0	0	0x0
GiveUp	0	0	0x0
PmtuAger	258658	258657	0x1B676275
DeadWait	0	0	0x0
Linger	0	0	0x0
ProcessQ	0	0	0x0

iss: 3572897128 snduna: 3572988441 sndnxt: 3572988441
irs: 815033843 rcvnxt: 815125275

sndwnd: 15833 scale: 0 maxrcvwnd: 16384
rcvwnd: 15681 scale: 0 delrcvwnd: 703

SRTT: 1000 ms, RTTO: 1003 ms, RTV: 3 ms, KRTT: 0 ms
minRTT: 0 ms, maxRTT: 1000 ms, ACK hold: 200 ms
uptime: 260336883 ms, Sent idletime: 2294 ms, Receive idletime: 2494 ms
Status Flags: active open
Option Flags: nagle, path mtu capable
IP Precedence value : 6

Datagrams (max data segment is 1460 bytes):

Rcvd: 9547 (out of order: 0), with data: 4785, total data bytes: 91431
Sent: 9550 (retransmit: 1, fastretransmit: 0, partialack: 0, Second Congestion: 0), with data: 4783, total data bytes: 91312

Packets received in fast path: 0, fast processed: 0, slow path: 0
fast lock acquisition failures: 0, slow path: 0
TCP Semaphore 0x7FAC28B5B0B8 FREE
R4#show run
Building configuration...

Current configuration : 2743 bytes
!
! Last configuration change at 22:07:00 UTC Mon Nov 17 2025
!
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname R4
!
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
address-family ipv4
exit-address-family
!
address-family ipv6
exit-address-family
!
!
no aaa new-model
!
!
!
!
!

!
!
!
!
!
!

!
!
!
!
!
!
!
!
!
!

ipv6 unicast-routing

subscriber templating
vtp domain cisco
vtp mode transparent

!
multilink bundle-name authenticated

!
!
!
!

license udi pid ISR4321/K9 sn FDO21491LXV
license accept end user agreement
license boot level securityk9

!
spanning-tree extend system-id

!
!

redundancy
mode none

[illegible]

```
!  
interface GigabitEthernet0/0/0  
ip address 192.168.5.1 255.255.255.0  
negotiation auto  
ipv6 address 2001:DB8:ACAD:5::1/64  
ipv6 ospf 10 area 0  
!  
interface GigabitEthernet0/0/1  
ip address 192.168.4.2 255.255.255.0  
ip ospf 1 area 0  
negotiation auto  
ipv6 address 2001:DB8:ACAD:4::2/64  
ipv6 ospf 10 area 0  
!  
interface Serial0/1/0  
no ip address  
shutdown  
!  
interface Serial0/1/1  
no ip address  
shutdown  
!  
interface GigabitEthernet0  
vrf forwarding Mgmt-intf  
no ip address  
shutdown  
negotiation auto  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
router-id 4.4.4.4  
redistribute connected subnets  
redistribute bgp 3 subnets  
redistribute isis level-2 subnets  
redistribute eigrp 10
```

```
!  
router isis  
!  
router bgp 3  
bgp consistency-checker error-message interval 5  
bgp log-neighbor-changes  
bgp graceful-restart restart-time 120  
bgp graceful-restart stalepath-time 360  
bgp graceful-restart  
bgp sso route-refresh-enable  
neighbor 2001:DB8:ACAD:5::2 remote-as 4  
neighbor 192.168.5.2 remote-as 4  
!  
address-family ipv4  
  redistribute connected  
  redistribute ospf 1 metric 10000 match internal external 1 external 2  
  redistribute isis level-2  
  redistribute eigrp 10  
  no neighbor 2001:DB8:ACAD:5::2 activate  
  neighbor 192.168.5.2 activate  
exit-address-family  
!  
address-family ipv6  
  redistribute connected  
  redistribute ospf 10 metric 10000 match internal external 2  
  redistribute eigrp 20  
  redistribute eigrp 10  
  redistribute ospf 1 metric 10000  
  neighbor 2001:DB8:ACAD:5::2 activate  
exit-address-family  
!  
ip forward-protocol nd  
no ip http server  
no ip http secure-server  
ip tftp source-interface GigabitEthernet0  
!  
!  
ipv6 router ospf 10
```

```
router-id 4.4.4.4
redistribute connected
redistribute bgp 3 metric 10000
```

```
!
!
!
!
```

```
control-plane
```

```
!
!
```

```
line con 0
```

```
stopbits 1
```

```
line aux 0
```

```
stopbits 1
```

```
line vty 0 4
```

```
login
```

```
!
!
```

```
End
```

```
Router 5:
```

```
R5#show run
```

```
Building configuration...
```

```
Current configuration : 2358 bytes
```

```
!
```

```
! Last configuration change at 21:44:48 UTC Mon Nov 17 2025
```

```
!
```

```
version 15.5
```

```
service timestamps debug datetime msec
```

```
service timestamps log datetime msec
```

```
no platform punt-keepalive disable-kernel-core
```

```
!
```

```
hostname R5
```

```
!
```

```
boot-start-marker
```

```
boot-end-marker
```

```
!
```

```
!  
vrf definition Mgmt-intf  
!  
address-family ipv4  
exit-address-family  
!  
address-family ipv6  
exit-address-family  
!  
!  
no aaa new-model  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!
```

```
!  
!  
!  
ipv6 unicast-routing  
!  
!  
!  
!  
!  
!  
!  
subscriber templating  
vtp domain cisco
```

```
vtp mode transparent
multilink bundle-name authenticated
!
!
!
!
license udi pid ISR4321/K9 sn FDO21400XZX
!
spanning-tree extend system-id
!
!
redundancy
mode none
!
!
vlan internal allocation policy ascending
!
!
!
!
!
!
interface Loopback0
ip address 192.168.10.1 255.255.255.0
ip router isis
!
interface GigabitEthernet0/0/0
ip address 192.168.6.1 255.255.255.0
ip router isis
negotiation auto
ipv6 address 2001:DB8:ACAD:6::1/64
ipv6 enable
ipv6 eigrp 20
!
interface GigabitEthernet0/0/1
ip address 192.168.5.2 255.255.255.0
ip router isis
negotiation auto
```

```
ipv6 address 2001:DB8:ACAD:5::2/64
ipv6 enable
ipv6 eigrp 20
!
interface Serial0/1/0
no ip address
shutdown
!
interface Serial0/1/1
no ip address
shutdown
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
interface Vlan1
no ip address
shutdown
!
router isis
net 49.0001.0000.0000.0001.00
is-type level-1
metric-style wide
redistribute connected
redistribute bgp 4 metric 10000 level-1-2
redistribute ospf 1
redistribute eigrp 10
!
router bgp 4
bgp log-neighbor-changes
neighbor 2001:DB8:ACAD:5::1 remote-as 3
neighbor 192.168.5.1 remote-as 3
!
address-family ipv4
    redistribute connected
```



```
redistribute isis level-1-2 metric 10000
redistribute ospf 1
redistribute eigrp 10
no neighbor 2001:DB8:ACAD:5::1 activate
neighbor 192.168.5.1 activate
exit-address-family
!
address-family ipv6
redistribute connected
redistribute eigrp 20
neighbor 2001:DB8:ACAD:5::1 activate
exit-address-family
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
ipv6 router eigrp 20
eigrp router-id 5.5.5.5
redistribute bgp 4 metric 10000 100 255 100 100
!
!
!
!
control-plane
!
!
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
!
!
end
```

R5#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

B 192.168.1.0/24 [20/10000] via 192.168.5.1, 02:42:31

B 192.168.2.0/24 [20/10000] via 192.168.5.1, 3d00h

B 192.168.3.0/24 [20/10000] via 192.168.5.1, 3d00h

B 192.168.4.0/24 [20/0] via 192.168.5.1, 3d00h

192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.5.0/24 is directly connected, GigabitEthernet0/0/1

L 192.168.5.2/32 is directly connected, GigabitEthernet0/0/1

192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.6.0/24 is directly connected, GigabitEthernet0/0/0

L 192.168.6.1/32 is directly connected, GigabitEthernet0/0/0

i L1 192.168.7.0/24 [115/20] via 192.168.6.2, 02:41:08, GigabitEthernet0/0/0

192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.10.0/24 is directly connected, Loopback0

L 192.168.10.1/32 is directly connected, Loopback0

i L1 192.168.11.0/24 [115/20] via 192.168.6.2, 3d00h, GigabitEthernet0/0/0

R5#show ipv6 route

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

B 2001:DB8:ACAD:1::/64 [20/10000]

via FE80::521C:B0FF:FE2D:7100, GigabitEthernet0/0/1

B 2001:DB8:ACAD:2::/64 [20/10000]
 via FE80::521C:B0FF:FE2D:7100, GigabitEthernet0/0/1

B 2001:DB8:ACAD:3::/64 [20/10000]
 via FE80::521C:B0FF:FE2D:7100, GigabitEthernet0/0/1

B 2001:DB8:ACAD:4::/64 [20/0]
 via FE80::521C:B0FF:FE2D:7100, GigabitEthernet0/0/1

C 2001:DB8:ACAD:5::/64 [0/0]
 via GigabitEthernet0/0/1, directly connected

L 2001:DB8:ACAD:5::2/128 [0/0]
 via GigabitEthernet0/0/1, receive

C 2001:DB8:ACAD:6::/64 [0/0]
 via GigabitEthernet0/0/0, directly connected

L 2001:DB8:ACAD:6::1/128 [0/0]
 via GigabitEthernet0/0/0, receive

D 2001:DB8:ACAD:7::/64 [90/3072]
 via FE80::521C:B0FF:FE2D:6801, GigabitEthernet0/0/0

L FF00::/8 [0/0]
 via Null0, receive

R5#show bgp neighbor

BGP neighbor is 192.168.5.1, remote AS 3, external link

BGP version 4, remote router ID 192.168.5.1

BGP state = Established, up for 3d00h

Last read 00:00:37, last write 00:00:24, hold time is 180, keepalive interval is 60 seconds

Neighbor sessions:

1 active, is not multisession capable (disabled)

Neighbor capabilities:

Route refresh: advertised and received(new)

Four-octets ASN Capability: advertised and received

Address family IPv4 Unicast: advertised and received

Enhanced Refresh Capability: advertised and received

Multisession Capability:

Stateful switchover support enabled: NO for session 1

Message statistics:

InQ depth is 0

OutQ depth is 0

Sent Rcvd

Opens:	1	1
Notifications:	0	0
Updates:	13	12
Keepalives:	4779	4776
Route Refresh:	0	1
Total:	4795	4792

Do log neighbor state changes (via global configuration)

Default minimum time between advertisement runs is 30 seconds

For address family: IPv4 Unicast

Session: 192.168.5.1

BGP table version 36, neighbor version 36/0

Output queue size : 0

Index 1, Advertise bit 0

1 update-group member

Slow-peer detection is disabled

Slow-peer split-update-group dynamic is disabled

	Sent	Rcvd
Prefix activity:	----	----
Prefixes Current:	5	5 (Consumes 600 bytes)
Prefixes Total:	16	13
Implicit Withdraw:	9	5
Explicit Withdraw:	2	3
Used as bestpath:	n/a	4
Used as multipath:	n/a	0

	Outbound	Inbound
Local Policy Denied Prefixes:	-----	-----
Bestpath from this peer:	11	n/a
Total:	11	0

Number of NLRIs in the update sent: max 3, min 0

Last detected as dynamic slow peer: never

Dynamic slow peer recovered: never

Refresh Epoch: 2

Last Sent Refresh Start-of-rib: 00:59:28

Last Sent Refresh End-of-rib: 00:59:28

Refresh-Out took 0 seconds

Last Received Refresh Start-of-rib: 00:59:29

Last Received Refresh End-of-rib: 00:59:29

Refresh-In took 0 seconds

	Sent	Rcvd
Refresh activity:	----	----
Refresh Start-of-RIB	1	1
Refresh End-of-RIB	1	1

Address tracking is enabled, the RIB does have a route to 192.168.5.1

Connections established 1; dropped 0

Last reset never

External BGP neighbor configured for connected checks (single-hop no-disable-connected-check)

Interface associated: GigabitEthernet0/0/1 (peering address in same link)

Transport(tcp) path-mtu-discovery is enabled

Graceful-Restart is disabled

SSO is disabled

Connection state is ESTAB, I/O status: 1, unread input bytes: 0

Connection is ECN Disabled, Minimum incoming TTL 0, Outgoing TTL 1

Local host: 192.168.5.2, Local port: 179

Foreign host: 192.168.5.1, Foreign port: 45641

Connection tableid (VRF): 0

Maximum output segment queue size: 50

Enqueued packets for retransmit: 0, input: 0 mis-ordered: 0 (0 bytes)

Event Timers (current time is 0x1B6CC11B):

Timer	Starts	Wakeups	Next
Retrans	4789	0	0x0
TimeWait	0	0	0x0
AckHold	4787	4702	0x0
SendWnd	0	0	0x0
KeepAlive	0	0	0x0
GiveUp	0	0	0x0
PmtuAger	0	0	0x0
DeadWait	0	0	0x0
Linger	0	0	0x0
ProcessQ	0	0	0x0

iss: 815033843 snduna: 815125389 sndnxt: 815125389

irs: 3572897128 rcvnxt: 3572988555

sndwnd: 15567 scale: 0 maxrcvwnd: 16384

rcvwnd: 15719 scale: 0 delrcvwnd: 665

SRTT: 1000 ms, RTTO: 1003 ms, RTV: 3 ms, KRTT: 0 ms

minRTT: 0 ms, maxRTT: 1000 ms, ACK hold: 200 ms

uptime: 260689049 ms, Sent idletime: 24579 ms, Receive idletime: 24378 ms

Status Flags: passive open, gen tcbs

Option Flags: nagle, path mtu capable

IP Precedence value : 6

Datagrams (max data segment is 1460 bytes):

Rcvd: 9562 (out of order: 0), with data: 4789, total data bytes: 91426

Sent: 9559 (retransmit: 0, fastretransmit: 0, partialack: 0, Second Congestion: 0), with data: 4791, total data bytes: 91545

Packets received in fast path: 0, fast processed: 0, slow path: 0

fast lock acquisition failures: 0, slow path: 0

TCP Semaphore 0x7F450DE1A6C0 FREE

R5#show ip bgp

BGP table version is 36, local router ID is 192.168.10.1

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 192.168.1.0	192.168.5.1	10000		0 3 ?	
*> 192.168.2.0	192.168.5.1	10000		0 3 ?	
*> 192.168.3.0	192.168.5.1	10000		0 3 ?	
*> 192.168.4.0	192.168.5.1	0		0 3 ?	
* 192.168.5.0	192.168.5.1	0		0 3 ?	
*> 0.0.0.0	0	32768		?	
*> 192.168.6.0	0.0.0.0	0		32768 ?	
*> 192.168.7.0	192.168.6.2	10000		32768 ?	

```
*> 192.168.10.0 0.0.0.0 0 32768 ?
*> 192.168.11.0 192.168.6.2 10000 32768 ?
```

Router 6:

R6#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

i L1 192.168.1.0/24

[115/10010] via 192.168.6.1, 02:42:36, GigabitEthernet0/0/1

i L1 192.168.2.0/24 [115/10010] via 192.168.6.1, 3d00h, GigabitEthernet0/0/1

i L1 192.168.3.0/24 [115/10010] via 192.168.6.1, 3d00h, GigabitEthernet0/0/1

i L1 192.168.4.0/24 [115/10010] via 192.168.6.1, 3d00h, GigabitEthernet0/0/1

i L1 192.168.5.0/24 [115/20] via 192.168.6.1, 3d00h, GigabitEthernet0/0/1

192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.6.0/24 is directly connected, GigabitEthernet0/0/1

L 192.168.6.2/32 is directly connected, GigabitEthernet0/0/1

192.168.7.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.7.0/24 is directly connected, GigabitEthernet0/0/0

L 192.168.7.1/32 is directly connected, GigabitEthernet0/0/0

i L1 192.168.10.0/24 [115/20] via 192.168.6.1, 3d00h, GigabitEthernet0/0/1

192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.11.0/24 is directly connected, Loopback0

L 192.168.11.10/32 is directly connected, Loopback0

R6#show ipv6 route

IPv6 Routing Table - default - 10 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
EX 2001:DB8:ACAD:1::/64 [170/281856]
 via FE80::6EDD:30FF:FEB6:6030, GigabitEthernet0/0/1
EX 2001:DB8:ACAD:2::/64 [170/281856]
 via FE80::6EDD:30FF:FEB6:6030, GigabitEthernet0/0/1
EX 2001:DB8:ACAD:3::/64 [170/281856]
 via FE80::6EDD:30FF:FEB6:6030, GigabitEthernet0/0/1
EX 2001:DB8:ACAD:4::/64 [170/281856]
 via FE80::6EDD:30FF:FEB6:6030, GigabitEthernet0/0/1
D 2001:DB8:ACAD:5::/64 [90/3072]
 via FE80::6EDD:30FF:FEB6:6030, GigabitEthernet0/0/1
C 2001:DB8:ACAD:6::/64 [0/0]
 via GigabitEthernet0/0/1, directly connected
L 2001:DB8:ACAD:6::2/128 [0/0]
 via GigabitEthernet0/0/1, receive
C 2001:DB8:ACAD:7::/64 [0/0]
 via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:7::1/128 [0/0]
 via GigabitEthernet0/0/0, receive
L FF00::/8 [0/0]
 via Null0, receive
R6#show run
Building configuration...

Current configuration : 1794 bytes
!
! Last configuration change at 20:50:30 UTC Mon Nov 17 2025
!
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
!
hostname R6
!


```
boot-start-marker
boot-end-marker
!
!
vrf definition Mgmt-intf
!
address-family ipv4
exit-address-family
!
address-family ipv6
exit-address-family
!
!
no aaa new-model
!
!
!
!
!
!
!
!
!
!
!

!
!
!
ipv6 unicast-routing
!
!
!
!
!
!
```

```
!  
subscriber templating  
vtp domain cisco  
vtp mode transparent  
multilink bundle-name authenticated  
!  
!  
!  
!  
license udi pid ISR4321/K9 sn FDO21491FHX  
!  
spanning-tree extend system-id  
!  
!  
redundancy  
mode none  
!  
!  
vlan internal allocation policy ascending  
!  
!  
!  
!  
!  
!  
interface Loopback0  
ip address 192.168.11.10 255.255.255.0  
ip router isis  
!  
interface GigabitEthernet0/0/0  
ip address 192.168.7.1 255.255.255.0  
ip router isis  
negotiation auto  
ipv6 address 2001:DB8:ACAD:7::1/64  
ipv6 enable  
ipv6 eigrp 20  
!  
interface GigabitEthernet0/0/1
```

```
ip address 192.168.6.2 255.255.255.0
ip router isis
negotiation auto
ipv6 address 2001:DB8:ACAD:6::2/64
ipv6 enable
ipv6 eigrp 20
!
interface GigabitEthernet0/1/0
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0/1/1
no ip address
shutdown
negotiation auto
!
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
!
interface Vlan1
no ip address
shutdown
!
router isis
net 49.0001.0000.0000.0002.00
is-type level-1
metric-style wide
!
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
!
!
```

```
ipv6 router eigrp 10
!
ipv6 router eigrp 20
eigrp router-id 6.6.6.6
!
!
!
!
control-plane
!
!
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
!
!
end
```

Problems

At first when trying to redistribute the routes across routers I came across a problem as I didn't have any BGP neighbors and when I was trying to find the cause of the problem, I was only looking at the BGP configurations and OSPF configurations but was not able to solve the problem. But when I looked at every configuration starting from the wiring then to the addressing, I noticed that the addressing on the interfaces was incorrect leading to the neighbor commands when configuring BGP not working therefore not establishing any neighbors. Before this problem I encountered a problem with redistributing the routes to other routers, even as I put in the basic redistribute commands the routes were not redistributing, after question marking my way through the CLI learning more about each command I realized that although entering a basic redistribute command without the metric may work for some protocols it did not work for others. For example, when I investigated my routing table, I had External Type 2 routes for EIGRP. This meant that when using the redistribute command I needed to specify the metric so that EIGRP can calculate the cost of the route to avoid routing loops, but also I need to specify the that I want to redistribute not the normal EIGRP routes but the External Type 2 routes as well. The last problem I encountered was with

making BGP share routes, just because I used the neighbor remote-as command didn't mean that I established a BGP session sharing all the routes I wanted to with the router. I went into the address-family level and found the neighbor activate command with the question mark which told the BGP process to share the routes within that address family to the other BGP neighbor.

Conclusion

This lab was extremely helpful in developing my understanding of external routing protocols. Taking the first step into learning the complexities of BGP and why it was become the most popular and most vital protocol when it comes to sustaining the internet and connectivity across the world will help me to understand how networks provide redundancy and reliable connections in future labs.